Evolution contributions

1 Genuine contributions

1.1 Group I

1.1.1 Row I

Only 4 ρ's

1.1.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.1}
$$

1.1.3 Row III

Only 3 ρ's

1.1.4 Row IV

Only 3 ρ's

1.1.5 Row V

We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.2}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.3}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x) \\
&- \frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.4}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite}\,\text{CT}\,NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{1.5}
$$

This is the counterterm of Part 2b: it is $-d^3N_{2b\,\text{UV}}$ term by term, so the two cancel. The label "finite" of the first d^3N_{2b} list is the right one; this one is not a finite part.

1.2 Group II

1.2.1 Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\Bigg] \qquad (1.6)
\end{aligned}
$$

1.2.2 Row II

Only 4 ρ's

1.2.3 Row III

Only 4 ρ's

1.2.4 Row IV

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\Bigg[\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x) \\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{1.7}
$$

1.3 Group III (with c.c.)

1.3.1 Row I

Only 3 ρ's

1.3.2 Row II

Only 3 ρ's

1.3.3 Row III

We write it in two parts:

1.3.4 First part:

Only 3 ρ's

1.3.5 Second part:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
&-4\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x) \\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\Bigg]
\end{aligned}
\tag{1.8}
$$

1.3.6 Row IV

Only 4 ρ's

1.3.7 Row V

Only 3 ρ's

1.3.8 Row VI

Only 3 ρ's

2 Symmetrizing three and four ρ's

2.1 From $4\,\rho' s$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x) \\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{2.1}
$$

Missing (M1): the two-ρ terms of `4rho.tex` ("Two ρ contribution", $dN^{(1)}$, $dN^{(2)}$ with c.c., with the fixes F1, F2 of `Final_evolution_check.pdf`). They belong to "From $4\,\rho$'s". Without them Secs. 1 and 2 do not add up to the JIMWLK

contribution of Sec. 3.

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x) \\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x) \\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x) \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x) \\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x) \\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
&+\frac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Bigg]
\end{aligned}
\tag{2.2}
$$

From three ρ's

2.2 Group I (with c.c.)

2.2.1 Row I

Only 4 ρ's

2.2.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+ 2\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x) \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x') \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.3}
$$

2.2.3 Row III

We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.4}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(x')\,\rho^l(x')\,\rho^h(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x') \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x') \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(x')\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.5}
$$

2.2.4 Row IV

We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{cb}(x')\,\rho^c(x')\,\rho^a(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{lc}(x)\,\rho^h(x')\,\rho^l(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{2.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{dc}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{2.7}
$$

2.2.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x') \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\Bigg]
\end{aligned}
\tag{2.8}
$$

2.3 Group II

2.3.1 Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{al}(w')\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^c(x)\,\rho^a(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^m(x)\,\rho^e(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^l(x)\,\rho^h(x') \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ha}(w)\,\rho^l(x')\,\rho^e(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(w)\,\rho^c(x)\,\rho^a(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^m(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^l(x)\,\rho^h(x') \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(w')\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x) \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x) \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ha}(w)\,\rho^m(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{2.9}
$$

2.3.2 Row II

Only 4 ρ's

2.3.3 Row III

Only 4 ρ's

2.3.4 Row IV

Only 2 ρ's

2.4 Group III (with c.c.)

2.4.1 Row I

We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{le}(x)\,U^{md}(w)\,U^{lb}(z)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(w)\,U^{hb}(z)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\Bigg]
\end{aligned}
\tag{2.10}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x') \\
+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^l(x)\,\rho^e(x') \\
- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^m(x)\,\rho^e(x') \\
- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^l(x)\,\rho^e(x') \\
+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^m(x)\,\rho^e(x') \\
- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^l(x)\,\rho^e(x') \\
+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^m(x)\,\rho^e(x') \\
- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(w)\,U^{hb}(z)\,U^{ma}(w')\,\rho^l(x)\,\rho^e(x') \\
- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.11}
$$

2.4.2 Row II

We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{al}(w)\,\rho^m(x')\,\rho^e(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{al}(w')\,\rho^m(x)\,\rho^e(x') \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(w')\,\rho^b(x)\,\rho^c(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{ad}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^c(x')\,\rho^e(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^m(x')\,\rho^e(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^c(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^m(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ca}(w')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w)\,\rho^l(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\Bigg]
\end{aligned}
\tag{2.12}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&+ \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w')\,\rho^e(x')\,\rho^l(x) \\
&- \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x') \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^e(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^e(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\Bigg]
\end{aligned}
\tag{2.13}
$$

2.4.3 Row III

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x') \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x') \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(x')\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{2.14}
$$

2.4.4 Row IV

Only 4 ρ's

2.4.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{ce}(w')\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x) \\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cd}(w)\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{2.15}
$$

2.4.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x) \\
&-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{eb}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x) \\
&+\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{eb}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x') \\
&-\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x) \\
&+\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x) \\
&+\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{2.16}
$$

3 JIMWLK contribution

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
-\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x) \\
+\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x') \\
-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x') \\
+\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(w')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x) \\
-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(w')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(x')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x) \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(x')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{3.1}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x) \\
&-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x) \\
&+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x) \\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x) \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{3.2}
$$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\Big[\\
+2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) - 2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\Big[\\
+2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - 2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x) - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) + 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
-2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}\Big[\\
-2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + 2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
-2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x) + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}\Big[\\
-2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + 2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x) + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\Big[\\
+2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + 2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
-2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x) - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\Big[\\
-2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x) \\
+2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\Big[\\
-2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
+2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big]\Bigg]
\end{aligned}
\tag{3.3}
$$

Color singlet form (projectile in a colour singlet: $\rho^b(x')\,\rho^c(x) \to \delta^{bc}\,\langle\rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$ with $x' \to y$; $K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $K(a,a,z) = \frac{1}{(a-z)^2}$; $U^\dagger = U^T$, any U)

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda}\right]\frac{1}{N_c^2-1}\int_{x,x'y,z,w,w'} e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\Bigg[\\
& \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[4N_c{\rm Tr}\{U^\dagger(w)U(w')\} \\
& - 2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\} - 2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- 4N_c{\rm Tr}\{U^\dagger(x)U(w')\} \\
& + 2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} + 2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[4N_c{\rm Tr}\{U^\dagger(x)U(y)\} \\
& - 2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\} - 2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- 4N_c{\rm Tr}\{U^\dagger(w)U(y)\} \\
& + 2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} + 2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[- 2N_c{\rm Tr}\{U^\dagger(w)U(w')\} + 2N_c{\rm Tr}\{U^\dagger(x)U(w')\} \\
& - {\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(w')U(z)\} - {\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\} \\
& + {\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\} + {\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w')\} \\
& + {\rm Tr}\{T^eU^\dagger(w')U(x)\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\} + {\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[- 2N_c{\rm Tr}\{U^\dagger(w)U(w')\} + 2N_c{\rm Tr}\{U^\dagger(w)U(y)\} \\
& + {\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w)U(z)\} + {\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\} \\
& - {\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(w)U(z)\} - {\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\} \\
& - {\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\} + {\rm Tr}\{T^eU^\dagger(w)U(y)\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[2N_c{\rm Tr}\{U^\dagger(x)U(w')\} - 2N_c{\rm Tr}\{U^\dagger(x)U(y)\} \\
& - {\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} - {\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\} \\
& + {\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(x)U(z)\} + {\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\} \\
& - {\rm Tr}\{T^eU^\dagger(w')U(x)\}\,{\rm Tr}\{T^eU^\dagger(x)U(z)\} - {\rm Tr}\{T^eU^\dagger(x)U(y)\}\,{\rm Tr}\{T^eU^\dagger(x)U(z)\}\Big] \\
& + \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[2N_c{\rm Tr}\{U^\dagger(w)U(y)\} - 2N_c{\rm Tr}\{U^\dagger(x)U(y)\} \\
& - {\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} - {\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\} \\
& + {\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\} + {\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(y)\} \\
& - {\rm Tr}\{T^eU^\dagger(w)U(y)\}\,{\rm Tr}\{T^eU^\dagger(y)U(z)\} + {\rm Tr}\{T^eU^\dagger(x)U(y)\}\,{\rm Tr}\{T^eU^\dagger(y)U(z)\}\Big]\Bigg]
\end{aligned}
\tag{3.4}
$$

where $K(y,y,z) = \frac{(y-z)\cdot(y-z)}{(y-z)^4}$

4 BFKL contribution

BFKL⊗LO is not part of the coefficient of $\log\frac{\vee}{\Lambda}$: that coefficient is the JIMWLK contribution of Sec. 3 alone (it is the sum of Secs. 1 and 2). BFKL⊗LO is the coefficient of $\log\vee$: it comes from the upper limits $p^+ = \vee$, $p^+ + k^+ = \vee$ of the rows, with $\log\frac{\vee}{k^+}$. The total is in Sec. 5.

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{\Lambda}\right]\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{4.1}
$$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{\Lambda}\right]\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) - 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
- 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) + 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\Big] \\
+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,N_c\,\rho^a(x)\,\rho^a(x') - f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') \\
- f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x') + f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x) \\
+ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') + f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)\Big] \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
- N_c\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\Big] \\
+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') - 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x') \\
- 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') + 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\Big[\\
- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x') \\
+ N_c\,U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') - N_c\,U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\Big] \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
+ 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) - 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x) \\
- 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\Big]\Bigg]
\end{aligned}
\tag{4.2}
$$

In color singlet form, we get (projectile in a colour singlet: $\rho^b(x')\,\rho^c(x) \to \delta^{bc}\,\langle\rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$ with $x' \to y$; $K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $K(a,a,z) = \frac{1}{(a-z)^2}$; $U^\dagger = U^T$, any U)

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{\Lambda}\Big]\log\Big[\frac{\vee}{k^+}\Big]\,\frac{1}{N_c^2-1}\int_{x,x'y,z,w,w'} e^{-ik\cdot(w'-w)}\langle\rho^f(x)\rho^f(y)\rangle\Bigg[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big] \\
+\,K(w,w',z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big] \\
+\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\Bigg]
\end{aligned}
\tag{4.3}
$$

where $K(x,y,z) = \frac{(x-z)\cdot(y-z)}{(x-z)^2\,(y-z)^2}$

5 Total

With your rule A, $\log\frac{\Lambda}{k^+} = -\log\frac{\vee}{\Lambda} + \log\frac{\vee}{k^+}$, and with $\mathcal{J}$, $\mathcal{B}$ the brackets $[\dots]$ of Secs. 3 and 4 (without the logarithm):

$$
\begin{aligned}
\frac{d^3N_{\rm LL}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\Big[\frac{\vee}{\Lambda}\Big]\,\mathcal{J} + \log\Big[\frac{\vee}{k^+}\Big](\mathcal{B}-\mathcal{J})\Big] \\
&= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\Big[\frac{k^+}{\Lambda}\Big]\,\mathcal{J} + \log\Big[\frac{\vee}{k^+}\Big]\,\mathcal{B}\Big].
\end{aligned}
\tag{5.1}
$$

At mid rapidity, $k^+ = \sqrt{\vee\Lambda}$, this is $\frac{1}{2}\log\frac{\vee}{\Lambda}\,(\mathcal{J}+\mathcal{B})$.